LITERATURE REVIEW

How manipulable are prediction markets?

Itzhak Rasooly and Roberto Rozzi

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ANALYTICAL ABSTRACT

The paper asks whether a trader who pushes the price of a prediction market away from its prior level can leave a lasting mark, or whether the behaviour of other traders erases the shock within hours. The authors build a model of risk-averse traders facing a constant-product automated market maker, and they predict partial reversion that slows over time, with more reversion in markets that have more traders, more activity and an outside source of probability estimates. They test it by placing randomised yes or no bets that move the price by five points in 817 markets on the Manifold platform, tracking hourly prices for 30 days and a snapshot at 60 days, and by repeating the design with a smaller real-money sample. Prices revert by about a quarter within a week, yet a gap of roughly seven points between treated groups survives for a month and remains clearly visible at 60 days. The heterogeneity patterns have the predicted sign but are only marginally significant, and the markets studied are very cheap to move, which limits what the result says about high-stakes markets.

1 Introduction

1.1 Paper identification and classification

The paper is Rasooly and Rozzi, *How manipulable are prediction markets?*, circulated as arXiv:2503.03312 in the subject class General Economics. It is a mixed paper: its Section 2 presents a theoretical model with three lemmas and a simulation exercise, and its Sections 4 to 6 report two pre-registered randomised field experiments. The field is experimental and financial economics, and the subfield is information aggregation and manipulation in prediction markets, with a direct connection to market efficiency and to the microstructure of automated market makers. The version read in full for this review is the complete HTML rendering of arXiv:2503.03312 (main text, theory proofs in Appendix A, tables and figures in Appendix B, robustness in Appendix C, and references). The text of the numbered footnotes could not be recovered from the rendering used, so statements that depend on footnotes are not made here. The original PDF could not be downloaded in this environment and is therefore not stored with this review.

1.2 Executive understanding

WHAT IS THE PAPER TRYING TO DISCOVER?

It tries to measure how long the effect of a single manipulative trade survives in a live prediction market, and which market characteristics make the effect shorter. The outcome is a price path, not a binary notion of manipulability.

WHY DOES THE QUESTION MATTER?

Prediction markets such as Polymarket, Kalshi and Manifold now handle large volumes and are cited as probability sources for elections and geopolitical events. If a small, cheap trade can move the quoted probability for weeks, the quoted number is a weaker public signal than assumed. The question is also an indirect test of the efficient markets hypothesis, because a price that carries only fundamentals should forget a random trade quickly.

WHAT WAS KNOWN BEFORE, AND WHAT IS MISSING?

The prevailing view, summarised by Wolfers and Zitzewitz (2004), is that manipulation has no discernible effect beyond a short transition. The causal evidence behind that view was thin: observational studies of historical manipulation lack a counterfactual price path, laboratory studies use a handful of markets, and the closest field design, Camerer (1998), used racetrack pari-mutuel markets where bets were cancelled after roughly twelve minutes. What was missing was a randomised, across-market field design with a long horizon and a model that predicts the shape of the post-shock price path.

WHAT DOES THE PAPER PROPOSE, HOW DOES IT INVESTIGATE, AND WHAT DOES IT FIND?

It proposes a disequilibrium model in which each trader optimises over a non-linear price schedule, and it randomises a five-point price shock across 817 markets. It finds persistence (a gap of 7.5 points after one week, 6.9 after 30 days and 5.6 after 60 days between yes and no groups, from an initial gap of 10 points), partial reversion that decelerates, and heterogeneity in the predicted direction.

2 Research Problem and Hypothesis

2.1 Research question

PAPER The paper asks: after an exogenous trade moves the price of a binary prediction market by five percentage points, what fraction of the price displacement is reversed by subsequent trading at horizons of one week, 30 days and 60 days, and does that fraction rise with the number of traders, with trading activity and with the availability of an external probability estimate?

2.2 Hypotheses

- Persistence: a manipulation has a non-zero effect on price in the long run, in the short run because adjustment takes trades and in the long run because traders learn from price.
- Partial reversion: because traders with beliefs on the other side of the new price trade against it, the price drifts back, quickly at first and then more slowly as the displacement shrinks.
- Heterogeneity: reversion is larger in markets with more traders, with higher activity (a faster effective clock) and with an external probability source (less learning from the market price).

2.3 Existing literature and research gap

PAPER The authors position the work against four strands: price-theory models of prediction markets (Manski 2006; Gjerstad 2005; Wolfers and Zitzewitz 2006), which assume price-taking and competitive equilibrium; historical studies of attempted manipulation (Rhode and Strumpf 2004, 2006; Hansen, Schmidt and Strobel 2004); laboratory experiments (Hanson, Oprea and Porter 2006; Veiga and Vorsatz 2009, 2010; Choo, Kaplan and Zultan 2022); and the racetrack field experiment of Camerer (1998). The stated gap is the absence of a model of the price path after a shock and of a well-powered field experiment at the market level.

CRITIQUE The gap is real for prediction markets. A related and older literature on permanent price impact in financial microstructure (Kyle 1985; Hasbrouck 1991) is not engaged, although it supplies the benchmark against which persistence should be judged, as discussed in Section 7.

2.4 Claimed contribution

CRITIQUE Two contributions are claimed: the first model of the post-manipulation price path, and the first large-scale field experiment on prediction market manipulation. The experimental contribution is demonstrated convincingly for the markets sampled, since randomisation is at the market level, the design is pre-registered and the sample is large. The theoretical contribution is demonstrated only in a weaker sense: the model yields qualitative predictions that are matched in sign by the data, but it is never calibrated or estimated, so the claim that the data validate the model rests on qualitative agreement.

3 Theoretical Framework and Mechanism

3.1 Market, traders and pricing

PAPER A single market trades yes shares (paying one unit if the event occurs) and no shares (paying one unit otherwise). Trader i has belief π_i that the event occurs, wealth w and a von Neumann-Morgenstern utility u with $u' > 0$, $u'' < 0$, $\lim_{w_s \to 0^+} u'(w_s) = \infty$ and decreasing absolute risk aversion. A trader buys either yes or no shares, never both, and maximises

$$\mathbb{E}[u] = \pi_i\, u\big(w + q - C(q)\big) + (1 - \pi_i)\, u\big(w - C(q)\big) \quad \text{(yes purchase of } q \text{ shares)}. \tag{1}$$

π_i	Subjective probability of the event held by trader i, in $[0, 1]$; beliefs differ across traders.
w	Initial wealth, common across traders, in currency units.
q	Number of yes shares purchased; each pays one unit in the state where the event occurs.
$C(q)$	Total cost of q shares, determined by the automated market maker.

Equation (1) says that the trader gets $w + q - C(q)$ if the event occurs and $w - C(q)$ otherwise. The shape of C is therefore what turns a belief into a quantity: with a flat price the problem is a Kelly-type betting problem (Kelly 1956), and with a rising price the trader internalises their own price impact.

The market maker holds reserves (y, n) of yes and no shares and applies the constant product rule. A trader who pays x currency units causes the maker to mint x yes and x no shares, so reserves become $(y + x,\ n + x)$, and the maker returns the number q of yes shares that restores the product:

$$(y + x - q)(n + x) = yn. \tag{2}$$

Solving for q gives

$$q = \frac{x\,(n + x + y)}{n + x}, \qquad C(q) = \frac{\sqrt{(n - q + y)^2 + 4nq} + q - n - y}{2}. \tag{3}$$

The first step follows from $y+x-q = yn/(n+x)$, hence $q = y+x-yn/(n+x) = \big[x(n+x)+y(n+x)-yn\big]/(n+x)$, which is the stated expression. The second expression inverts the first for $x = C(q)$ by solving a

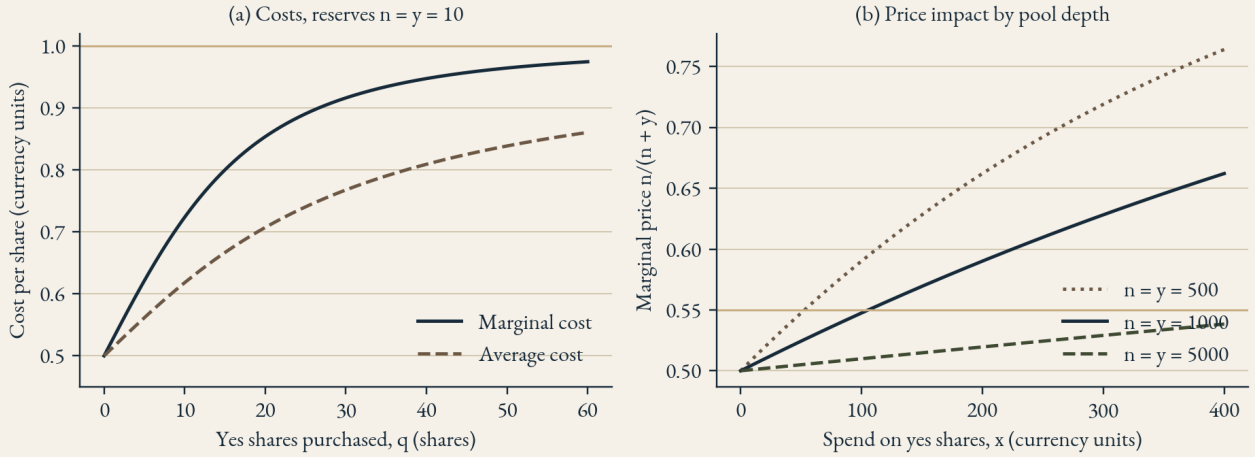
quadratic in x. Differentiating gives the marginal cost and its slope (Appendix A of the paper):

$$MC(q) = \frac{1}{2}\left(\frac{n+q-y}{\sqrt{(n-q+y)^2+4nq}}+1\right), \qquad MC'(q) = \frac{2ny}{\left((n-q+y)^2+4nq\right)^{3/2}} > 0. \quad (4)$$

y, n	Reserves of yes and no shares held by the automated market maker.
$MC(q)$	Price of the q-th marginal yes share.
$p = n/(n+y)$	Marginal price at zero quantity, which is the quoted probability.

Substituting $q = 0$ in Equation (4) gives $MC(0) = n/(n+y) = p$, and letting $q \to \infty$ gives $MC \to 1$, because the expression under the root is dominated by $(n+q-y)^2$. Hence Lemma 1: marginal and average cost start at p, rise strictly, and approach one. Figure 1(a) plots the two schedules for a pool of ten shares per side. The average-cost limit follows from L'Hôpital's rule, and its monotonicity from signing a derivative that reduces to $nq^2y > 0$ after squaring both sides.

Figure 1. Cost schedules and price impact under the constant product rule



Source: generated by the reviewer from Equations (2) to (4); panel (a) uses reserves of ten shares per side as in Figure A1 of the paper, panel (b) uses balanced pools of 500, 1000 and 5000 shares. The horizontal line in (b) marks a price of 0.55.

INFERENCE A closed form for the price impact of a trade follows from Equation (2). After spending x on yes shares, reserves are $(yn/(n+x),\ n+x)$, so the marginal price is $p' = (n+x)^2/[(n+x)^2+yn]$. Setting $p' = 0.55$ in a balanced pool $n = y$ gives $(n+x)/n = \sqrt{0.55/0.45}$, hence $x = 0.1055\,n$. With $n = y = 1000$ the experimenter pays about 105.5 currency units and receives about 201 yes shares. Cost is proportional to pool depth, which is why the cap of 200 Mana on the cost of a five-point move (roughly two dollars at the platform rate of 100 Mana per dollar) selects very shallow markets.

3.2 Optimal trading and the role of the marginal price

PAPER Lemma 2 states that, with $p = n/(n+y)$, a trader with $\pi_i > p$ buys a positive quantity of yes shares, a trader with $\pi_i = p$ holds nothing and a trader with $\pi_i < p$ buys no shares. The proof first shows that buying the wrong side lowers expected wealth: if $\pi_i > p$ and the trader buys $q_n > 0$ no shares, then, since the average cost of no shares exceeds $y/(n+y)$,

$$\mathbb{E}[W] < w + q_n \left(\frac{n}{n+y} - \pi_i \right) < w, \tag{5}$$

and a risk-averse agent never accepts lower expected wealth with added risk. It then differentiates Equation (1) at $q_y = 0$ using $C'(0) = p$:

$$\left. \frac{\partial\, \mathbb{E}[u]}{\partial q_y} \right|_{q_y=0} = u'(w)\,(\pi_i - p) > 0. \tag{6}$$

The derivative is positive when $\pi_i > p$, so staying out is not optimal, and existence of an optimum on a compact budget set gives a strictly positive quantity. Equation (6) is the extension of Arrow (1965): even a risk-averse trader is locally risk neutral, so any positive expected edge, however small, justifies a bet.

PAPER Lemma 3 gives the comparative static that drives reversion. For an interior optimum, the first-order condition is

$$\frac{\pi_i\, u'(w + q - C(q))}{(1-\pi_i)\, u'(w - C(q))} = \frac{C'(q)}{1 - C'(q)}. \tag{7}$$

Left side Ratio of expected marginal utilities in the event and no-event states, the trader's shadow price of the bet.

Right side Price odds, $C'(q)/(1-C'(q))$, which rise with the quantity bought.

Equation (7) says that the trader buys until the odds the market charges equal the odds implied by belief and marginal utility. A manipulation raises n and lowers y with $k = ny$ fixed. The paper applies the implicit function theorem to $f(q,n) = g(q,n)$, where f and g are the two sides of Equation (7), obtaining

$$\frac{dq}{dn} = \frac{\partial g/\partial n - \partial f/\partial n}{\partial f/\partial q - \partial g/\partial q} < 0. \tag{8}$$

The signs are as follows. The shock raises marginal cost, so $\partial g/\partial n > 0$ (the cost-slope derivative in the paper is positive). It raises $C(q)$, which lowers wealth in both states, and under decreasing absolute risk aversion the ratio of marginal utilities $u'(w+q-C)/u'(w-C)$ falls, so $\partial f/\partial n < 0$. Concavity gives $\partial f/\partial q < 0$ and Lemma 1 gives $\partial g/\partial q > 0$. The numerator of Equation (8) is positive and the denominator negative, so $dq/dn < 0$: every trader on the yes side holds fewer yes shares after the price rises, and traders with beliefs between p and $p + \Delta$ switch sides. That aggregate response pushes the

price back, which is the logic of reversion. The result is local and qualitative. It says nothing about how far or how fast the price returns.

3.3 Simulation: speed and extent of reversion

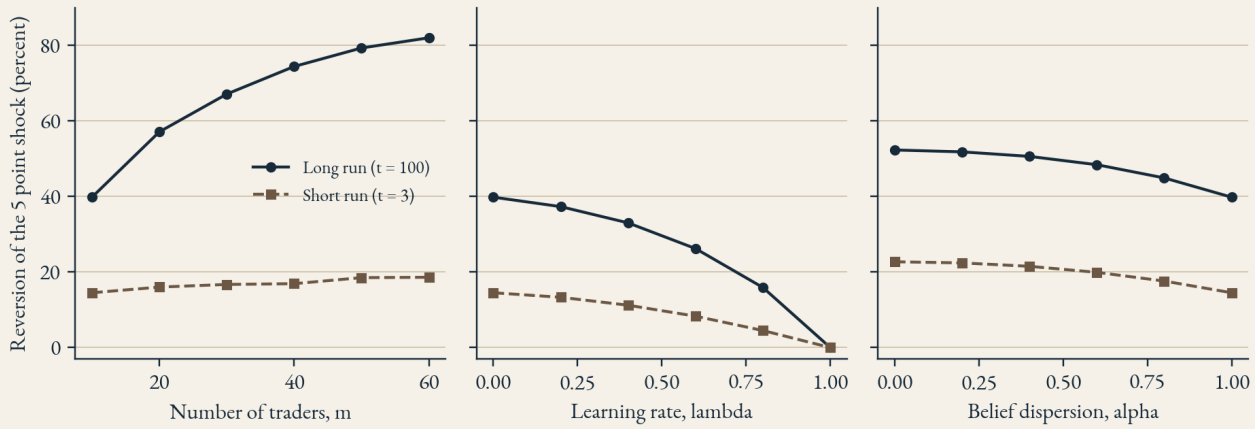
PAPER To obtain quantities the authors assume log utility, for which the optimal spend is explicit, and simulate 10^7 optimisation problems. In each period one randomly chosen trader re-optimises. Baseline parameters are $m + 1 = 11$ traders with wealth $w = 100$, beliefs $\pi_i = i/m$, reserves $(n, y) = (1000, 1000)$, and 100 periods before and after a five-point shock. Beliefs may respond to price through the learning rule of Manski (2006),

$$\pi_i' = \lambda p + (1 - \lambda)\,\pi_i, \qquad \pi_i = \alpha \frac{i}{m} + (1 - \alpha)\,0.5, \tag{9}$$

- λ Learning rate: weight a trader places on the market price when forming a posterior; $\lambda = 0$ means beliefs ignore price.
- α Belief dispersion: $\alpha = 1$ gives uniform beliefs on $[0, 1]$, $\alpha = 0$ gives a common belief of 0.5.
- m Number of traders beyond the first, hence total trader wealth $(m + 1)w$.

The baseline simulation reverts only about 40 percent of the shock in the long run. The paper's Table A1 gives short-run (three periods) and long-run (100 periods) reversion, and Figure 2 plots it.

Figure 2. **Model-implied reversion of a five-point manipulation under alternative market conditions**



Source: values copied from Table A1 of the paper (official source: the authors' simulations of 10,000 runs each); the figure itself was redrawn by the reviewer. Reversion is the fraction of the initial shock undone, short run at period 3 and long run at period 100.

INFERENCE Equation (9) has a sharp implication at the boundary. At $\lambda = 1$ every posterior equals the current price, so by Lemma 2 every trader is indifferent at any price and demand is zero: there is a continuum of resting prices and the manipulated price stays put forever (Table A1 shows exactly zero reversion). The price process is then a martingale in which the shock is permanent, which is the benchmark in Kyle-type models where trades are interpreted as information. Persistence in the data can therefore be a sign of rational learning, not of inefficiency, a point developed in the critique. Table A1 also shows that the model reaches 82 percent long-run reversion at 60 traders but only 40 percent at ten, so with a median market of about 27 traders at the time of the bet, matching the

observed 41 to 44 percent reversion at 60 days requires a sizeable learning rate. The paper does not make this calibration explicit.

3.4 Generalisation to Maniswap

PAPER Manifold uses Maniswap, which holds $y^p n^{1-p}$ constant, with $p \in (0, 1)$ the initial price. At $p = 0.5$ it reduces to the constant product rule. The authors state without proof that modified versions of the three lemmas hold. This is plausible but unverified in the text read.

3.5 Closed form for log utility

CRITIQUE The paper reports an explicit expression for the optimal spend under log utility (its Equation 6). The radical signs of that expression were lost in the rendering available, and a reconstructed version did not reproduce the optimum, so it is not reproduced here. As an independent check, the reviewer solved the log-utility problem numerically from Equation (7): for $n = y = 1000$, $w = 100$ and $\pi_i = 0.7$ the optimal spend is 37.3 currency units (73.3 shares), and for $\pi_i = 0.9$ it is 77.8 (149.9 shares). These figures are generated by the reviewer, not taken from the paper.

4 Data and Empirical Strategy

4.1 Setting and data

PAPER The experiment ran on Manifold Markets, then the largest prediction market website by number of markets, with about 10,000 active users. Most markets are user-created and user-resolved, a large share of volume comes from bots, and trading is in a platform currency (Mana) convertible to charity donations at 100 Mana per dollar. The sample comprises 817 markets selected by five exclusions: resolution by the end of 2025 on an external event, at least ten traders at the time of the bet, a cost of at most 200 Mana to move the price five points either way, at least seven days of market life and 30 days to resolution, and no market closely related to one already in the sample. The authors collected hourly prices from 24 hours before to 30 days after the bet (about 600,000 observations) and one snapshot at 60 days, together with trader count, trading volume, 24 hour volume, comments and an indicator for a duplicate question on Metaculus.

Table 1. **Experiments, samples and measurement horizons**

Experiment	Shock and groups	Horizon and frequency	Markets
Main, Mana markets (Dec 2023 to Apr 2024)	Yes bet (up 5 points), no bet (down 5), control	Hourly for 30 days, snapshot at 60 days	817
Follow-up, Sweepcash markets (Dec 2024 to Feb 2025)	Yes bet (up 4 points), no bet (down 4); no control	Daily for six or seven days	110

Source: copied from Sections 4 and 6 of the paper (official). In the follow-up, 89 markets have seven days of data because Manifold discontinued Sweepcash markets on 13 February 2025.

Table 2. **Descriptive statistics of the main sample (n = 817)**

Variable	Mean	Std. dev.	Minimum	Maximum
24 hour volume (Mana)	24.7	119.9	0	1,701
Total volume (Mana)	1,879	4,279	89	56,485
On Metaculus (indicator)	0.093	0.291	0	1
Comments	3.53	7.26	0	83
Traders at time of bet	27.0	25.0	10	300

Source: Table 1 of the paper (official). Total volume of 1,879 Mana corresponds to about 19 dollars at the platform rate, a computation by the reviewer.

4.2 Empirical specification and identification

PAPER Each market is independently assigned to yes, no or control with equal probability. The primary comparison is between the yes and no groups, the best-powered contrast. For each horizon t the authors estimate a cross-sectional regression across markets,

$$p_{t,i} = \beta_0 + \beta_1 \mathbb{1}_{Y,i} + \beta_2 \mathbb{1}_{C,i} + \beta_3 p_{-1,i} + u_i. \tag{10}$$

$p_{t,i}$ Price of market i, t hours after the bet, in probability units.

$\mathbb{1}_{Y,i}$, $\mathbb{1}_{C,i}$ Indicators for the yes group and the control group; the no group is the omitted category.

$p_{-1,i}$ Price in the hour before the bet, a baseline control added to increase precision.

β_1 Yes minus no difference in mean price at horizon t; equals about 0.10 at $t = 0$ by construction.

Because assignment is random, $\hat{\beta}_1$ has a causal reading as the differential effect of a yes versus a no manipulation. Reversion at horizon t is then summarised as $1 - \hat{\beta}_1(t)/0.10$. Symmetry of the two shocks corresponds to $\beta_1 = 2\beta_2$, tested with an F test ($p = 0.124$, not rejected). Heterogeneity uses median splits of each moderator x_i and the interacted regression

$$p_{167,i} = \beta_0 + \beta_1 \mathbb{1}_{Y,i} + \beta_2 \mathbb{1}_{C,i} + \beta_3 p_{-1,i} + \beta_4 \mathbb{1}_{Y,i} x_i + \beta_5 \mathbb{1}_{C,i} x_i + \beta_6 p_{-1,i} x_i + u_i, \tag{11}$$

in which β_4 is the difference in treatment effect between above-median and below-median markets. The follow-up uses Equation (10) without a control group, with daily horizons. All standard errors are heteroskedasticity-robust, computed horizon by horizon.

EVIDENCE The identifying variation is the random assignment of the shock across markets. Each market contributes one observation per horizon, which removes the serial dependence that would otherwise contaminate inference within a market. The cost is that the confidence bands plotted over hours are pointwise, not simultaneous.

4.3 Data limitations and biases

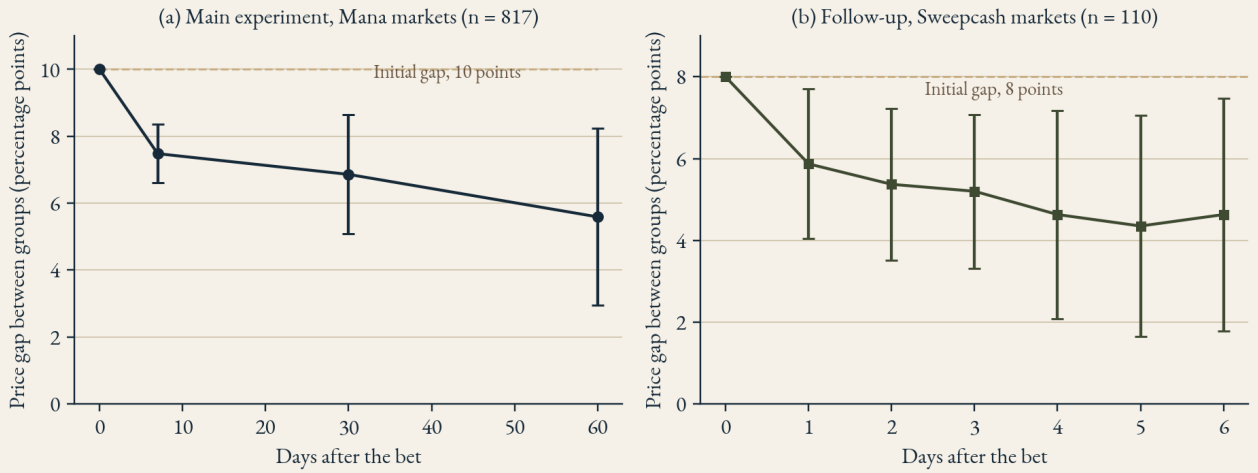
- Selection of thin markets: the cost cap and the ten-trader floor yield markets with mean total volume of about 19 dollars. External validity to deep markets is not established, as the authors acknowledge.
- Post-treatment conditioning: the 60 day analysis drops markets that resolved before 60 days (46 of 817) to exclude mechanical convergence to zero or one. Resolution timing is determined after assignment, although the chance that it depends on the bet is small.
- Possible detection of the experimenter and the platform's bot activity: the paper does not report whether the bets were traceable to a recognisable account, which matters for the learning channel.
- Spillovers across markets are addressed with a language-model screen and a theoretical argument (Appendix C); the screen is not reproducible in a strict sense because it depends on a particular model and prompt.

5 Results

5.1 Persistence and reversion of the manipulation

EVIDENCE Figure 3(a) summarises the main results from the paper's tables. In the regression of Equation (10) the coefficient on the yes group is 0.0748 with a standard error of 0.00446 at one week, 0.0686 (0.00908) at 30 days and 0.0559 (0.0135) at 60 days for the full sample, all with $p < 0.01$. Relative to the initial gap of 0.10, this is 25 percent reversion after a week, 31 percent after 30 days and 44 percent at 60 days. The 60 day figure includes markets that had already resolved; for the 771 unresolved markets the coefficient is 0.0586 (0.0114), which is 41 percent reversion. The t statistics are about 17 at one week, 7.6 at 30 days and 4.1 at 60 days in the full sample. In intuitive terms, a bet that cost at most two dollars displaced the quoted probability by a gap of about 7 points a month later, against a 10 point initial gap.

Figure 3. Price gap between the yes and no groups after a manipulation, by horizon



Source: coefficients and robust standard errors copied from Tables 2, 3 and A5 of the paper (official), redrawn by the reviewer; bars show 95 percent confidence intervals (1.96 times the standard error). Panel (a) 60 day point is the full-sample estimate and includes resolved markets; the initial gap of 10 points is by design. Panel (b) day 7 is omitted because it uses a smaller sample.

Table 3. Yes versus no effect at one week, 30 days and 60 days (regression of Equation 10)

Quantity	1 week	30 days	60 days, all	60 days, unresolved
Yes coefficient, $\hat{\beta}_1$	0.0748	0.0686	0.0559	0.0586
Robust standard error	0.00446	0.00908	0.0135	0.0114
Coefficient on $p_{-1,i}$	1.000	1.028	1.020	1.035
R^2	0.933	0.798	0.617	0.723
Markets	817	817	817	771
Implied reversion (reviewer)	25 percent	31 percent	44 percent	41 percent

Source: Tables 2 and 3 of the paper (official), column with only the baseline price as control; implied reversion is $1 - \hat{\beta}_1/0.10$, computed by the reviewer. All coefficients have $p < 0.01$.

EVIDENCE Table 3 answers the question of what survives and for how long. The dependent variable is the market price at the horizon, the key regressor is the yes indicator, and the columns change the horizon and, in the last two, whether resolved markets are included. The coefficient on $p_{-1,i}$ is close to one, which means the baseline price carries through almost one for one and explains most of the variation (R^2 falls from 0.93 to 0.62 as noise accumulates). The pattern of reversion is decelerating: 25 percent in the first week, then six further points over the next three weeks, consistent with the model, which predicts slowing as the displacement shrinks. The table does not establish that the persistent gap is a distortion rather than information, nor does it reveal what drives the heterogeneity within the single average.

EVIDENCE The control group allows a second check. The no group differs from control by β_2, and the paper reports $\beta_1 = 2\beta_2$ is not rejected ($p = 0.124$) and the control-versus-no comparison shows a similar sustained gap (Figure A5), so yes and no manipulations are reasonably symmetric. For the follow-up in Sweepcash markets the gap begins at 8 points (a four-point shock each way) and falls to 5.9 after one day, 4.4 after five days and 4.6 after six days, all with $p < 0.01$ in the full sample (Table A5; Figure 3(b)). In the 89-market subsample with seven days the day-seven gap is 3.5 points ($p < 0.05$).

5.2 Heterogeneity

EVIDENCE Table 4 reproduces the one-week treatment effects by median split of each moderator, from the regression of Equation (11).

Table 4. **Heterogeneity of the one week effect by market type (median splits)**

Moderator	Above median	Below median	Difference	p value	Reversion above	Reversion below
On Metaculus	0.053	0.077	0.024	0.096	47 percent	23 percent
24 hour volume	0.049	0.081	0.032	0.041	51 percent	19 percent
Total volume	0.069	0.081	0.012	0.191	31 percent	19 percent
Comments	0.069	0.084	0.015	0.071	31 percent	16 percent
Total traders	0.067	0.083	0.017	0.056	33 percent	17 percent

Source: Table 4 of the paper (official); the reversion columns are $1-\beta/0.10$ computed by the reviewer. All treatment effects in the first two columns have $p<0.01$. For Metaculus the split is the binary indicator, about 9 percent of the sample.

EVIDENCE Every moderator has the predicted sign. Markets with an external probability source show 47 percent reversion against 23 percent for the rest, and markets with an above-median trader count show 33 percent against 17 percent. The only difference significant at the five percent level is 24 hour volume. The Metaculus contrast is significant only at ten percent and total volume is not significant ($p=0.191$). The finer splits in Tables A3 and A4 show a monotone pattern: for markets in the top quartile by trader count (about 56 traders) reversion rises to 37 percent, yet the treatment effect remains significant at one percent. The results therefore support the direction of the comparative statics but not their precision.

6 Robustness and Additional Tests

Table 5. **Robustness tests: concern, test, result and assessment**

Concern	Test reported	Result	Does it resolve the concern?
Baseline imbalance or low power	Add all baseline variables as controls (Table 3)	Coefficient 0.0748 becomes 0.0759 at one week; 0.0686 becomes 0.0697 at 30 days; reversion 25 percent becomes 24 percent	Yes. Randomisation makes controls unnecessary and the stability confirms balance.
Cross-market spillovers (arbitrage and information)	Theoretical argument that random yes and no assignment cancels spillovers in expectation; drop markets flagged as related by a language model (Table A6)	Yes coefficient 0.075 in both the main and the reduced sample (556 and 482 markets, control excluded)	Largely. The argument is sound for symmetric assignment, and dropping the most related markets leaves the estimate unchanged, but the screen is a subjective filter.

Concern	Test reported	Result	Does it resolve the concern?
Resolution mechanically ending the effect	Exclude markets resolved by day 60 (Table 2)	Coefficient 0.0559 becomes 0.0586; no effect among resolved markets (n=46, large standard error)	Yes for the question asked, with the post-treatment caveat noted above.
Mana is not real money	Sweepcash follow-up on dollar-denominated markets (Table A5)	Gap 8 points falls to 4.6 after six days; effect positive each day	Partly. Same direction, but only 110 markets, six to seven days and a different, more active market type.
Time horizon and power	Extend from 7 to 30 and 60 days	Significant at one percent at all horizons; standard errors grow from 0.0045 to 0.0135	Partly. The 60 day test is a post hoc extension, not pre-registered.
Symmetry of yes and no shocks	F test of $\beta_1 = 2\beta_2$	$p = 0.124$, not rejected	Partly. Failure to reject is weak evidence of equality with an unknown power.

Source: Sections 5 and 6 and Appendix C of the paper (official results); the assessments are the reviewer's.

CRITIQUE The robustness battery addresses internal validity well but leaves the larger questions of external validity and interpretation untouched. None of the tests addresses whether the persistent gap reflects belief change by informed traders, a sticky reference price, or inattention. The tests also do not correct for the number of horizons, moderators and specifications examined.

7 Critical Evaluation

7.1 What the paper establishes

ESTABLISHED

Small randomised trades move the quoted price of cheap Manifold markets by amounts that remain statistically and economically visible after one week and after 30 days, with partial reversion that is quick at first and slow afterwards. The result survives control variables and a spillover screen.

SUPPORTED BUT CONDITIONAL

The effect remains visible at 60 days, which depends on the extension beyond the pre-registered horizon and on how resolved markets are handled. The follow-up on dollar-denominated markets shows the same sign, but its size and horizon limit the comparison with the main experiment.

SUGGESTIVE

Markets with more traders, higher activity and an external probability source revert more. The signs match the model, but three of five contrasts are significant only at ten percent or not at all, the splits are not randomised, and the moderators are correlated.

> **NOT ESTABLISHED**
>
> That the model explains the data quantitatively, that persistence indicates market inefficiency, that prediction markets are manipulable at the stakes where manipulation would matter, and that persistent price displacement reduces forecast accuracy, since no outcome-based accuracy test is reported.

7.2 Strengths

STRENGTHS RECOGNISED BY THE AUTHORS

- Randomisation at the market level with 817 units, which removes the counterfactual problem that afflicts historical studies and gives the power for heterogeneity analysis.
- A pre-registered design with a power calculation (849 markets for 90 percent power at three points), followed by an out-of-sample follow-up with a pre-registered sample size.
- A model whose predictions are specified before the data are examined, and a real-money robustness experiment.

STRENGTHS IDENTIFIED INDEPENDENTLY

- The estimand is simple and the identifying variation is transparent: one number per market per horizon, so standard errors are not distorted by within-market autocorrelation.
- The three lemmas are proved in an appendix and the proofs are checkable: the constant product algebra and the first-order condition reproduce, which allows an independent numerical check of optimal trades.
- The treatment cost is small and bounded (at most 200 Mana per market), which makes the experiment cheap to replicate and extend, and the authors disclose a change of shock size from five to four points in the follow-up with its reason.
- The paper separates a manipulation estimate from a platform-specific incentive discussion, and documents a donation response to a platform devaluation to argue that Mana is not play money.

7.3 Weaknesses and critique

SINGLE MOST IMPORTANT WEAKNESS

Problem. Persistence is interpreted as manipulability and as a test of efficiency, but a persistent displacement is exactly what a rational, efficient market produces when it cannot tell a noise trade from an informative one. **Why it matters.** In Kyle (1985) and in the empirical decomposition of Hasbrouck (1991), trade innovations move prices permanently because traders and market makers update on order flow. A market that fully reverted a shock would be reading every trade as noise, which is itself inefficient. The paper's own learning parameter λ in Equation (9) is this channel, and at $\lambda = 1$ persistence is complete and the market is efficient in the semi-strong sense

given its information set. **Potential bias.** The inference from persistence to inefficiency, and the claim that persistent shocks threaten accuracy, are not identified by this experiment, because the experimenter is observationally an informed trader. **How it could be tested.** Link persistence to resolution outcomes: if the shock carries no information, resolution frequencies should not differ by treatment arm conditional on the pre-bet price, and calibration of treated versus control prices should be compared at resolution. The sample of 817 markets limits power for this test, but a larger replication on resolved markets would settle it.

OTHER WEAKNESSES

- Thin markets: the mean total volume is about 19 dollars, and the median is not reported. *Why it matters:* the result may only show that cheap markets are cheap to move. Under Equation (3) cost scales with pool depth, so the five-point shock in a market with 100 times the liquidity costs about 100 times as much, as the authors note. *Potential bias:* overstatement of manipulability for markets that matter. *Test:* stratify by depth, or use a dose-response design that scales the shock to depth.
- No structural link between model and data: the simulation uses 11 traders, 100 periods and a random re-optimisation rule with no mapping between a period and an hour. *Why it matters:* the claim that the model predicts the shape of the price path cannot be falsified, because any process in which the gap decays at a falling rate matches the qualitative pattern. *Test:* estimate λ, α and an effective trader count by simulated method of moments on the observed reversion path.
- Multiple testing and median splits: five moderators and several splits, with p values of 0.041 to 0.191 and none adjusted. *Why it matters:* only one of the five contrasts is nominally significant at five percent. Applying the Benjamini and Hochberg (1995) step-up rule to the five p values (0.041, 0.056, 0.071, 0.096, 0.191), as the reviewer did, rejects none at a false discovery rate of 0.05 and all five only at a rate of 0.20. *Potential bias:* overreading of heterogeneity. *Test:* adjusted p values and randomisation inference as in Young (2019).
- Endogenous moderators: trader count, volume and comments correlate with market age, topic and attention, none of which is randomised. *Why it matters:* heterogeneity estimates are descriptive. *Test:* a joint regression with all moderators and market age as controls.
- Pointwise confidence bands: the figures over 720 hours present 95 percent bands at each horizon. *Why it matters:* statements about the shape of reversion over time need simultaneous bands or a parametric path. *Test:* a sup-t band or a fit of a decay model to the whole path.
- Authors' disclosed risk of being noticed: the working paper circulated during the follow-up, and the authors changed the shock size to avoid suspicion. *Why it matters:* traders who suspect an experiment may trade against it. *Potential bias:* faster reversion in the follow-up.

8 Contradictions and Edge Cases

CRITIQUE Under what conditions should the result disappear? Three cases can be reasoned out from the model. First, at very small λ with many traders the model reverts more than 80 percent (Table A1), so the persistent gap in the data points to a high learning rate or a small effective number of traders. A market with a hundred active, wealthy traders should show much less persistence than this sample, and the paper's own median split on trader count shows only a 16-point difference in reversion between halves. Second, the shock of five points is both large enough to detect and small enough to be plausible. The authors conjecture a U-shaped relation between shock size and persistence. The model suggests why: the marginal trader who disagrees with a very large displacement faces a large and cheap profit opportunity, since by Equation (6) the local expected gain per share for a trader whose belief lies on the other side of the displaced price is proportional to the gap between belief and price, and that gap grows with the displacement. Reversion should therefore accelerate for large shocks even without any credibility consideration.

CRITIQUE Extreme values. As $q \to \infty$ marginal cost tends to one, so large manipulations become uneconomical for the manipulator as well, which sets a natural upper bound. At the opposite extreme, markets priced near zero or one move little in probability terms for a given spend, because dp'/dx shrinks with depth near the boundaries, and the paper does not report whether the results depend on the pre-bet price. The regression of Equation (10) imposes a common slope on $p_{-1,i}$, so a nonlinear dependence of reversion on the starting price would be absorbed by the control and not by the treatment coefficient.

CRITIQUE A worked illustration. In a balanced pool of 1000 shares per side, the five-point shock costs about 105.5 currency units (see Section 3), and a trader with $\pi_i = 0.7$ facing a pre-shock price of 0.5 would spend about 37.3 units (computed with log utility and wealth 100). If the shock raises the marginal price to 0.55, that same trader's optimal spend falls, in the direction of Lemma 3, but remains positive because $0.7 > 0.55$. Each such trader's own bet is a fraction of the experimenter's 105.5 units, so undoing the shock requires several similarly capitalised optimists to be active in the same market, and thin participation or limited attention, not irrationality, can suffice for persistence. This is the reviewer's illustration, not a result in the paper.

CRITIQUE Alternative mechanisms for the same data. First, anchoring: traders on a platform that displays only the current price and a chart may treat the displayed probability as a reference point. Second, attention: reversion requires someone to look at a market, and the median market with 24 hour volume near zero may go unobserved for days. Third, bots, which the paper mentions as exploiting inefficiencies, may either revert shocks quickly or amplify them if they follow price momentum. The experiment cannot separate these mechanisms because it randomises the shock, not the channel.

CRITIQUE Outside the original setting. The paper itself reports an informal survey in which the user base skews male, American and aligned with rationalist and effective altruism communities, motivated partly by charity and reputation, and the pricing mechanism (Maniswap) differs from order book exchanges. An order book with resting limit orders, as on Polymarket or Kalshi, can absorb a given trade at depth that differs from a pool by construction, so the translation of cost to price impact must be redone market by market.

9 Further Literature

Table 6. Literature connections

Paper	Shared idea	Relationship	Why it matters
Wolfers and Zitzewitz (2004)	Prediction markets aggregate information; manipulation is transient	Contradicted: the field experiment finds effects 60 days later	Sets the benchmark belief that the paper challenges.
Camerer (1998)	Randomised bets in a betting market to test manipulation	Preceded; different pricing rule, bets cancelled after minutes	Direct experimental precedent, with opposite findings on persistence.
Rhode and Strumpf (2004, 2006)	Historical manipulation of political stock markets	Preceded; observational, no counterfactual	Shows that manipulation is attempted; the paper supplies the causal estimate.
Manski (2006); Gjerstad (2005); Wolfers and Zitzewitz (2006)	Price as a function of belief distribution and risk aversion	Theoretical foundation, extended to non-linear pricing and disequilibrium	Provides the learning rule and the interpretation of price as a probability.
Hanson and Oprea (2009)	A manipulator can raise accuracy by increasing the return to information	Contradicts the view that manipulation necessarily harms accuracy	Suggests the welfare effect of persistence is ambiguous.
Kyle (1985)	Price impact of order flow with an informed trader and noise traders	Related benchmark for permanent price impact	Frames persistence as rational learning; see the review in this library.
Hasbrouck (1991)	Permanent price impact of trade innovations	Related empirical decomposition	Gives a measure that the experiment could be recast in.
Allen and Gale (1992)	Trade-based manipulation in equilibrium	Theory of why manipulation can be profitable	Explains why persistence need not be eliminated by arbitrage.
Fama (1970); Grossman and Stiglitz (1980)	Efficient markets and the cost of information	Foundation of the efficiency interpretation	Persistence is evidence on efficiency only under a model of what traders can infer.
Choo, Kaplan and Zultan (2022)	Manipulation and trust in prediction markets	Laboratory complement	Laboratory study of manipulation and trust, relevant to the learning channel.
Ottaviani and Sorensen (2015)	Price reaction with heterogeneous beliefs and wealth effects	Related theory of reversal and momentum	A route to explaining short-run reversal and long-run drift.
Angeris and Chitra (2020)	Constant function market makers and price oracles	Related mechanism design	Characterises the pricing family to which the constant product rule belongs.

Source: reviewer's classification; works marked as preceding or foundational are cited in the paper itself, while Hasbrouck, Kyle, Ottaviani and Sorensen, and Angeris and Chitra are added by the reviewer.

10 Research Gaps

10.1 Paper-level gaps

- **Gap:** the model is never calibrated. *Why it matters:* without estimated parameters the model is untestable beyond signs. *Existing limitation:* simulation periods have no calendar mapping. *Way forward:* estimate λ, α and an effective trader count by simulated method of moments.
- **Gap:** no outcome-based test of whether persistent shocks harm accuracy. *Why it matters:* welfare consequences depend on it. *Existing limitation:* only 127 of 817 markets had resolved when the paper was written. *Way forward:* follow all markets to resolution and compare Brier scores by treatment arm.
- **Gap:** a single shock size and one trade per market. *Why it matters:* optimal manipulation is unknown. *Existing limitation:* the authors state this. *Way forward:* a dose-response design.

10.2 Literature-level gaps

- **Gap:** no common framework linking price impact in prediction markets to the permanent-and-transitory decomposition in equity markets. *Why it matters:* it would let results be compared across market types. *Existing limitation:* different literatures. *Way forward:* estimate impact decay curves for prediction markets with the vector autoregression of Hasbrouck (1991).
- **Gap:** the effect of manipulation in high-stakes markets is unknown. *Why it matters:* policy decisions use these markets. *Existing limitation:* experiments are costly there. *Way forward:* natural experiments from large documented trades.

10.3 Methodological gaps

- **Gap:** median splits discard information and invite multiple testing. *Why it matters:* heterogeneity claims are fragile. *Existing limitation:* the pre-registration prescribes them. *Way forward:* continuous moderators in a causal forest or a pre-registered joint model with adjusted inference.
- **Gap:** pointwise bands for a path. *Why it matters:* claims about the shape of reversion. *Existing limitation:* the paper estimates each horizon separately. *Way forward:* a panel with market-clustered errors and simultaneous bands.

11 Research Opportunities

Idea 1. Structural estimation of learning and belief dispersion from reversion paths

RESEARCH QUESTION. What learning rate, belief dispersion and effective trader count rationalise the observed reversion path of 25 percent at one week and 41 percent at 60 days, and what do they imply for persistence in deeper markets?

MOTIVATION. The model is only qualitatively confronted with data, and the learning rate is the parameter that separates manipulation from information.

HYPOTHESIS. The data require a learning rate well above zero; the estimated learning rate falls with the number of traders and with the presence of an external source.

ECONOMIC MECHANISM. Traders update beliefs toward the price, so a shock partly becomes accepted information.

LITERATURE CONNECTION. Manski (2006); Hasbrouck (1991); Kyle (1985).

DATA. The authors' replication package and public Manifold trade histories.

VARIABLES. Group-level price paths, trader counts, volume, comments, bot share of volume.

METHOD. Simulated method of moments matching treatment effects at several horizons by market type, with the model of Section 3 re-implemented with explicit clock time.

IDENTIFICATION STRATEGY. Randomised treatment supplies the moments; heterogeneity across market types identifies the parameters.

EXPECTED CONTRIBUTION. A calibrated model that can predict persistence for markets of any depth.

MAIN RISK. Weak identification: several parameter combinations may produce similar paths.

Idea 2. Dose-response design with shocks scaled to market depth

RESEARCH QUESTION. How does the persistence of a manipulation vary with its size relative to market depth, and is the relationship U-shaped as the authors conjecture?

MOTIVATION. The optimal manipulation is unknown and determines the real threat.

HYPOTHESIS. Persistence rises with shock size up to a credibility threshold and then falls.

ECONOMIC MECHANISM. Larger shocks carry more apparent information, but also create larger profit opportunities for traders who disbelieve them.

LITERATURE CONNECTION. Kyle (1985); Allen and Gale (1992); this paper.

DATA. New Manifold markets and, where possible, dollar-denominated markets on another platform.

VARIABLES. Price at several horizons, shock size as a fraction of liquidity, market covariates.

METHOD. Randomise each market to one of four or five shock sizes and a control, in the pre-registered style of the original.

IDENTIFICATION STRATEGY. Random assignment of dose.

EXPECTED CONTRIBUTION. A response curve instead of a single point.

MAIN RISK. Cost grows with dose; a large dose is also more detectable by other traders.

Idea 3. Do persistent shocks distort calibration?

RESEARCH QUESTION. Conditional on the pre-bet price, does the treated market price predict resolution differently from the control price?

MOTIVATION. Persistence matters for welfare only if it reduces accuracy; this is the paper's missing outcome test.

HYPOTHESIS. Under pure noise, resolution frequencies do not depend on treatment, so treated prices are mis-calibrated by the size of the surviving gap.

ECONOMIC MECHANISM. A displaced price remains a worse forecast until information arrives; a manipulator that is mistaken for an informed trader reduces accuracy.

LITERATURE CONNECTION. Hanson and Oprea (2009); Fama (1970); Grossman and Stiglitz (1980).

DATA. Resolved markets from the original sample and the replication package.

VARIABLES. Resolution outcome, price at a fixed time before resolution, treatment indicator, Brier score.

METHOD. Regress the outcome on the price and on the treatment indicator; compare Brier scores.

IDENTIFICATION STRATEGY. Random assignment.

EXPECTED CONTRIBUTION. First direct measure of the accuracy cost of manipulation.

MAIN RISK. Low power: with about 270 markets per arm, the standard error of a resolution-rate difference is near 0.04, comparable to the effect sought.

Idea 4. The role of bots and attention in reversion

RESEARCH QUESTION. Does the speed of reversion depend on the share of volume traded by automated accounts and on the recent frequency of human trades?

MOTIVATION. Reversion requires someone to trade; bots may supply or distort it.

HYPOTHESIS. Markets with a higher pre-bet bot share revert faster, and the effect of the bot share is larger than that of the trader count.

ECONOMIC MECHANISM. Automated traders arbitrage deviations from a model price, acting as the unwinding force that a thin human crowd cannot supply.

LITERATURE CONNECTION. Rothschild and Sethi (2016) on trading strategies in a prediction market; Hasbrouck (1991).

DATA. Manifold trade and user data, with bot flags if the interface exposes them, joined to the experimental markets.

VARIABLES. Bot share of volume in the pre-period, number of distinct traders in the first 24 hours, price paths.

METHOD. Interact the treatment indicator with the pre-period moderators using continuous splines; adjust for multiple testing.

IDENTIFICATION STRATEGY. Random treatment with pre-determined moderators.

EXPECTED CONTRIBUTION. A mechanism for the heterogeneity that the paper only documents.

MAIN RISK. The moderator is not randomised and may proxy for market quality.

Idea 5. Impact decay in a real-money exchange, linked to illiquidity

RESEARCH QUESTION. In real-money binary markets with an order book, how does the persistent component of price impact vary with measured illiquidity?

MOTIVATION. It connects the prediction market evidence to the permanent and temporary impact decomposition in microstructure, and to the Amihud-type illiquidity measure used elsewhere in this library.

HYPOTHESIS. The persistent share of impact is higher in less liquid contracts, and the half-life of the transitory part is shorter in contracts with more frequent trading.

ECONOMIC MECHANISM. Order flow is interpreted as information in proportion to the market's uncertainty about the identity of the trader.

LITERATURE CONNECTION. Kyle (1985); Hasbrouck (1991); Amihud (2002); this paper.

DATA. Public trade and quote history from a regulated event-contract exchange or a crypto-based venue.

VARIABLES. Signed trade size, mid-price paths at minute, hour and day horizons, depth, spread, volume.

METHOD. Event study of the largest trades with a Hasbrouck-style vector autoregression, comparing impact decay with illiquidity across contracts.

IDENTIFICATION STRATEGY. Observational; use trade-size shocks relative to depth, with wallet or account history to separate informed from uninformed large trades where identifiers are public.

EXPECTED CONTRIBUTION. A bridge between the experiment and standard microstructure estimates for deep markets.

MAIN RISK. Endogeneity of large trades; randomised intervention would be preferable but raises compliance and exchange-rule issues that need legal review.

11.1 Ranking

Table 7. **Ranking of research ideas (scores from 1 to 5, assigned by the reviewer)**

Idea	Novelty	Feasibility	Data	Method	Importance	Contribution	Total
1. Structural estimation	4	4	4	4	4	5	25
2. Dose-response design	4	3	4	5	5	5	26
3. Calibration and accuracy	4	3	4	4	5	4	24
4. Bots and attention	3	4	3	3	3	3	19
5. Impact decay in real money	4	3	3	3	4	4	21

Source: reviewer's subjective scoring.

The three strongest ideas are the dose-response design, the structural estimation and the calibration test. The dose-response experiment answers the paper's own stated open question with the cleanest identification, and extends it to a quantity (persistence as a function of size) that is directly relevant to policy. The structural estimation is the cheapest to run, because the data already exist, and it

turns a qualitative model into one that predicts for deeper markets. The calibration test addresses the welfare question the paper leaves open, although its power is limited unless the sample is enlarged.

12 Learning Framework

12.1 Prerequisite concepts

- Expected utility, risk aversion and the Arrow-Pratt measure of absolute risk aversion, and Kelly betting.
- Automated market makers and the constant product invariant, and the relation between reserves and price.
- Randomised experiments, robust standard errors, pre-registration and multiple hypothesis testing.
- The efficient markets hypothesis and information-based price impact (Kyle 1985).

12.2 Core concepts learned

- A persistent price displacement is compatible with efficiency if prices cannot distinguish noise from information; persistence is thus an upper bound on manipulability and not a measure of inefficiency.
- Under a constant product rule, cost to move the price scales with pool depth, and optimal demand is pinned down by the marginal price at zero quantity.
- Reversion is partial and decelerating, and its extent depends on how many traders hold beliefs on the other side and how strongly they learn from price.

12.3 Important equations

Equation (2), the constant product invariant, determines how any purchase moves the price. Equation (6), $u'(w)(\pi_i - p)$, states that a trader bets if and only if belief differs from price. Equation (7) gives the optimal bet against rising costs. Equation (9) is the learning rule that controls persistence. Equation (10) is the experimental estimating equation, and the reversion measure $1 - \hat{\beta}_1/0.10$ converts it to an intuitive quantity.

12.4 Methods learned

Market-level randomisation with an outcome measured repeatedly over time, estimated as a sequence of cross-sectional regressions with a pre-treatment control for precision; median-split heterogeneity analysis; simulation of agent decisions with explicit optimisation; and randomisation-based arguments for the absence of spillovers when treatment is symmetric.

12.5 Questions to test understanding

COMPREHENSION

1. What happens to the marginal price after a purchase of x yes shares, and how much does a five-point move cost in a balanced pool of 1000 shares?
2. Why is the coefficient on $\mathbb{1}_{Y,i}$ about 0.10 at $t = 0$, and how is reversion computed from it?
3. What does the paper find about the size of the effect after 60 days, and why are resolved markets treated separately?

ANALYSIS

1. Why does a risk-averse trader bet whenever $\pi_i \neq p$, however small the difference? (Use Equation 6.)
2. Why does assigning yes and no shocks with equal probability neutralise cross-market spillovers in Equation (10)?
3. Why does the model give no reversion at $\lambda = 1$, and what is the equivalent statement in a Kyle-type model?

RESEARCH

1. What experiment would separate persistence caused by learning from persistence caused by inattention?
2. How would you design a test of whether persistence reduces forecast accuracy, and what sample would it need?
3. How would the result change on an order book exchange with resting liquidity at several price levels?

12.6 Papers to read next

Hasbrouck (1991); Kyle (1985); Hanson and Oprea (2009); Camerer (1998); Ottaviani and Sorensen (2015); Angeris and Chitra (2020).

13 Conclusion

13.1 One-sentence thesis

Small randomised trades leave a statistically clear and economically sizeable mark on cheap prediction market prices for at least 60 days, with partial reversion that slows over time and varies with market type as a simple model of risk-averse traders predicts in sign.

13.2 Three most important insights

- Persistence and reversion coexist: about a quarter of the shock is undone within a week, the rest decays slowly, and about 41 percent has reverted at 60 days.
- Cost to manipulate scales with pool depth, so the sample's results apply first of all to shallow markets and cannot be assumed for deep ones.
- Persistence is not by itself evidence of inefficiency, since a market that learns from order flow will treat a noise trade as partly informative.

13.3 Strongest evidence

The one-week and 30 day estimates in Table 3: a gap of 7.5 and 6.9 points with t statistics of 17 and 7.6 from random assignment across 817 markets, stable to controls and to the removal of related markets.

13.4 Biggest weakness

The interpretation of persistence as manipulability or inefficiency, since the experimenter's trade is observationally indistinguishable from an informed trade and no accuracy outcome is tested.

13.5 Most important research gap

Whether persistent displacement reduces the accuracy of forecasts, and whether results extend to deep, high-stakes markets.

13.6 Best research opportunity

A dose-response experiment that scales shocks to market depth and follows markets to resolution, which tests the paper's conjecture about optimal manipulation and the accuracy cost in a single design.

13.7 Three papers to read next

Hasbrouck (1991); Kyle (1985); Hanson and Oprea (2009).

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